Lab: Case Study Analysis — Learning, Plasticity, and Clinical Change

Objective

Analyze clinical and experimental cases that illustrate learning as generative model updating, connecting synaptic mechanisms to cognitive change.

Part 1: Critical Period Plasticity

Case: Hubel and Wiesel's classic monocular deprivation experiment: kittens raised with one eye sutured shut during the critical period develop permanent blindness in that eye, even when the suture is removed. Kittens deprived after the critical period close show normal vision recovery.

- Interpret this in Active Inference terms: what happens to the generative model's visual component during critical period deprivation?
- Why is the effect irreversible after the critical period? (Hint: precision on existing priors increases, closing the window of plasticity.)
- What does this tell us about the importance of early experience for model architecture?

Write your response here

Part 2: Sleep and Memory Consolidation

Case: Participants learn a list of word pairs (e.g., "dog-umbrella") before either a period of sleep or an equivalent period of wakefulness. Sleep group shows 20% better recall and, crucially, better ability to generalize to related word pairs they never saw.

- How does NREM sleep replay contribute to parameter consolidation (strengthening correct associations)?
- How might Bayesian Model Reduction during sleep explain the generalization benefit (extracting the abstract structure from specific examples)?
- Predict what would happen if participants were deprived specifically of NREM sleep (e.g., through targeted auditory disruptions).

Write your response here

Part 3: PTSD and Maladaptive Priors

Case: Veteran J.B., 34, experiences flashbacks, hypervigilance, and startle responses 2 years after combat deployment. A car backfiring triggers a full fight-or-flight response. J.B. knows intellectually that he is safe but cannot override the automatic response.

- Interpret PTSD as the creation of high-precision priors about environmental danger. Why does intellectual knowledge of safety fail to override these priors?
- How does prolonged exposure therapy work in Active Inference terms? (Hint: repeated exposure to trauma cues without negative outcome generates prediction errors that gradually reduce the precision of danger priors.)
- What is the role of precision in the therapeutic process? What might "resistance to treatment" look like in precision terms?

Write your response here

Part 4: Expertise and Perceptual Learning

Case: Radiologists with 10+ years of experience can detect breast cancer on mammograms within 200 milliseconds — faster than conscious recognition. Novice radiology trainees require several seconds and make significantly more errors.

- How does the expert's generative model differ from the novice's? (Consider precision, hierarchical depth, and automated policy selection.)
- How does perceptual learning (improving sensitivity to subtle features through practice) reshape the early visual processing hierarchy?
- Is the expert's 200ms detection "conscious"? What does Active Inference predict about the relationship between speed and awareness?

Write your response here

Part 5: Synthesis

Write a 200-word synthesis explaining how these four cases collectively demonstrate that learning is not information storage but active, precision-weighted reorganization of the generative model.

Write your response here

Lab Summary

Part	Skill Developed	Case Type
1	Developmental reasoning	Critical period plasticity
2	Sleep research analysis	Memory consolidation and BMR
3	Therapeutic reasoning	PTSD and maladaptive priors
4	Expertise analysis	Perceptual learning in radiology
5	Integrative synthesis	Connecting learning cases to model updating